If you look at the final line of output, you'll see the output we expected from our Ruby script: "hello". So somehow it worked. But how? Why?

The docker run command has the following format:

```
$ docker run [OPTIONS] <image> <command>
```

This command starts a new container based on <image>, and executes <command> inside the container. You may find it helpful to think about it in two parts: docker run [OPTIONS] <image> says what type of container we're going to run, whereas <command> says what we want to run inside the container.

So looking back at our command, we have:

```
1 docker run ruby:2.6
2 ruby -e "puts :hello"
```

The first part says that we want to run a container based on the ruby:2.6 image. As we said previously (Before We Begin, on page 6), an image is a bundled-up package of everything needed to run (a specific) container. The ruby:2.6 image is no exception; it has Ruby 2.6 preinstalled, with all its dependencies, allowing us to create containers capable of running this version of Ruby.

The second part of the command specifies *what* we want to run inside the container. In this case, we're saying we want to run the Ruby interpreter with a script passed in using the command-line option -e. The script is the most basic you can imagine: it simply outputs the word "hello".

Terminology: Running an Image



We may occasionally talk of *running* an image, but strictly speaking, this is incorrect. Images can't be run directly; they are immutable factories for creating containers. Instead, what we mean is that we *create a container based on the image*, and it's the container that can be run.

Our docker run command will work on any machine that has Docker installed—even one without Ruby.

How is that possible? It's all very well that the image ruby:2.6 has Ruby installed, but how did we magically have it on our computer?

In actual fact, we didn't.

When we executed the docker run command, you may have noticed that it said Unable to find image 'ruby:2.6' locally. Docker then proceeded to download the ruby:2.6